AIRTEL SYNC HACKATHON 2026 · TRACK A SUBMISSION

PROJECT UDYAM DIGITAL

Accelerating India's 63M+ MSMEs from Inquiry to Active Fibre & Cloud in One Sitting (<24 Hours)

₹2,450

Target CAC

-83.7% from ₹15,000 baseline

< 24 Hours

Lead-to-Active

-95% reduction from 17.6 days

28.0%

Conversion Rate

3.8x uplift from 7.36% baseline

0 Calls

Status Chases

100% self-serve order visibility

🔗 SUBMISSION DELIVERABLES & INTERACTIVE LINKS

🎥 **Product Walkthrough Video (Google Drive)**

▶ Click here to view the end-to-end video demonstration

💻 **Working Full-Stack Web Application**

Built on Flask, Multi-Lingual Web Speech Voice, 12-Specimen Smart KYC, RF ML CRM

IN The MSME Market Opportunity

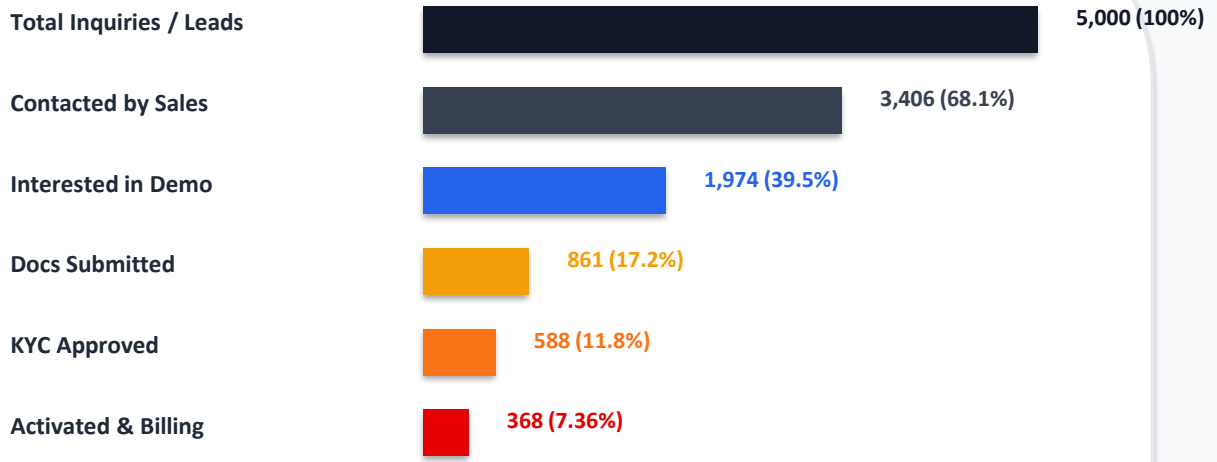
- **63M+ Micro, Small & Medium Enterprises**
  Backbone of Indian GDP (30%) and employment (110M+ workers). Only ~15% currently possess managed enterprise digital infrastructure.

- **₹2.5 Lakh Crore Connectivity & SaaS TAM**
  Rapidly shifting from unorganized consumer broadband to bundled B2B connectivity, cyber safety, cloud storage, and POS systems.

- **Airtel's Strategic Advantage**
  Unmatched nationwide fibre footprint, 5G standalone edge, trusted enterprise relationship, and comprehensive business product ecosystem (25+ B2B products).

- **The Growth Bottleneck**
  Massive top-funnel demand is throttled by complex paper onboarding, multi-visit document chases, and non-transparent pricing.

⚠️ Current Funnel Reality (smb_leads.csv Baseline)



- **17.6 Days Latency:** Physical form submissions & manual back-office verifications
- **₹15,000 High CAC:** Multiple field RM visits, paper courier, high churn leakage
- **3.0 Status Chases:** Frustrated SMB owners calling reps repeatedly for updates
- **11.2% Early Churn:** Missed SLA expectations, fragmented multi-vendor setups

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🏪 Micro Retail
55% of Leads

₹1,998 / mo
10% Bundle Discount

Target Persona: Kiranas, apparel shops, local pharmacies, food joints

📦 **Included Solutions:**

✓ Airtel Biz Fibre 100 Mbps (Unlimited)

✓ Smart Cloud POS & Soundbox

✓ Secure Airtel Guest Wi-Fi Portal

✓ Digital Payment Settlement Integration

💡 **Business Impact:**
Fast checkout, customer Wi-Fi marketing, zero downtime for UPI payments.

💼 Professional Offices
25% of Leads

₹2,698 / mo
12% Bundle Discount

Target Persona: Chartered Accountants, law firms, doctors, consultants

📦 **Included Solutions:**

✓ Airtel Biz Fibre 300 Mbps High Speed

✓ Static IP Address (for ERP & Tally)

✓ Airtel Cloud Backup (1 TB Encrypted)

✓ DNS Cyber Threat Protection Shield

💡 **Business Impact:**
Remote client data access, compliant automated daily backups, secure audits.

🏬 Small Multi-Site
15% of Leads

₹3,798 / mo
15% Bundle Discount

Target Persona: Retail chains, diagnostic labs, boutique gym branches

📦 **Included Solutions:**

✓ Airtel Biz Fibre 300 Mbps Primary

✓ Automated 4G/5G Wireless Failover SIM

✓ Airtel Cloud CCTV (8-Channel, 30d retention)

✓ Centralized SD-WAN Hub Management

💡 **Business Impact:**
100% operational uptime, centralized surveillance, unified multisite billing.

🏭 Small Manufacturers
5% of Leads

₹6,900 / mo
18% Bundle Discount

Target Persona: Auto-component workshops, fabrication, food packaging

📦 **Included Solutions:**

✓ 1:1 Dedicated Internet Leased Line (10 Mbps)

✓ Industrial IoT Machine Telemetry Gateway

✓ Next-Gen Hardware Firewall (NGFW)

✓ Priority 24/7 Enterprise Mission SLA

💡 **Business Impact:**
Guaranteed low latency, machine telemetry logging, perimeter cyber security.

📹 Click to Watch Demo Video (Google Drive)

🏗️ 4-Tier Scalable Solution Architecture

⚡ Presentation Layer
Responsive HTML5/JS Web App, Airtel Theme, Web Speech API (Multilingual Voice in EN/HI/TA)

⚡ Application & API Gateway
Modular Flask REST Framework, Advisor Engine, KYC Verification Service, Order Lifecycle Tracker

⚡ Intelligence & Decision Layer
GenAI Rule Engine, Random Forest Propensity Classifier (AUC 0.645, 12 features), 12-Check KYC Engine

⚡ Data & Compliance Vault
Mock GSTIN & Udyam Registries (6,200+ records), DPDP Act 2023 Consent Audit Logs, AES-256 Storage

🚀 One-Sitting Customer Journey (< 10 Minutes)

Step 1: Voice / Text Needs Discovery
SMB owner speaks in Hindi, Tamil, or English; AI matches archetype & formulates customized proposal in <7 seconds.

Step 2: Instant Transparent Quote
Auto-applies bundle discounts (10–18%), computes 18% GST itemization, and outputs branded PDF summary & WhatsApp quote.

Step 3: Instant Registry Pre-fill
Live lookup against GSTIN & Udyam registries pre-fills enterprise name, constitution, and authorized signatories without manual entry.

Step 4: Smart Specimen KYC Verification
Rule-based document validation checks blur, skew, 90-day recency, Mod-36 checksum, and signatory authority in seconds.

Step 5: Order Dispatch & Real-Time Tracking
Generates Order ID, dispatches hardware, allocates field installation engineer, and provides customer self-serve tracker.

📹 Click to Watch Demo Video (Google Drive)

Core Innovations: GenAI Advisor & Smart KYC Engine

GenAI Multilingual Advisor

✓ Voice-First Consultation (Web Speech API)
Micro-entrepreneurs often prefer speaking over typing. Supports real-time speech-to-text in English, Hindi, and Tamil.

✓ Archetype-Driven Semantic Engine
Maps informal prompts (e.g., 'mujhe payment machine aur fast internet chahiye') into structured B2B SKUs in <7 seconds.

✓ Multi-Lingual Business Rationales
Generates personalized, non-technical value rationales explaining expected ROI in the customer's native tongue.

✓ One-Click Bundle Hand-Off
Recommended bundle instantly populates Quote Builder with exact SKUs, contract tenure, and promotional pricing.

Smart KYC Pipeline & DPDP Compliance

Verified Against All 12 Hackathon KYC Specimen Files (K01–K12):
- K03 (Pvt Ltd) & K12 (NE Proprietorship) APPROVED (94% & 91% Trust Score)
- K01, K02, K04–K11 accurately flagged & REJECTED with specific compliance causes

🔒 Algorithmic Verification
Mod-36 GSTIN checksum validation, PAN entity code checking, Laplacien variance blur detection, Hough skew detection.

🔒 Registry Cross-Verification
Real-time automated reconciliation against mock GSTIN (3,534 records) & Udyam (2,723 records) registries.

🔒 Signatory Governance
Validates board resolutions, MOA signatory authority, partner consents, and power-of-attorney documentation.

🔒 DPDP Act 2023 Compliance
AES-256 encryption at rest, TLS 1.3 in transit, explicit consent capture, purpose limitation, and full audit trails.

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🎯 Partner CRM & ML Propensity Engine

Trained Random Forest Classifier (AUC 0.645)
Trained on smb_leads.csv (5,000 leads) utilizing 12 engineered features: business sector, employee size, inquiry channel, engagement velocity, and city circle.

3-Tier Propensity Scoring
Categorizes every incoming inquiry into High, Medium, or Low Propensity with estimated probability of activation within 30 days.

Automated Next-Best-Action (NBA) Prescriptions
High Propensity: Priority RM phone call within 2 hours with bundled discount.
Medium Propensity: Automated WhatsApp catalog & customer case study.
Low Propensity: Nurture email sequence with educational video guides.

Sales Channel Enablement
Provides field channel partners with real-time conversion insights, pending document alerts, and commission visibility.

🗒️ Quote Builder & Real-Time Order Tracker

✓ Full 25-Product Enterprise Catalog
Connectivity, Cloud SaaS, Cyber Security, IoT, and Collaboration SKUs clearly priced with no hidden charges.

✓ Automated GST & Bundle Discount Engine
Instant calculation of 18% GST (CGST + SGST / IGST) and applied volume savings so the customer sees the exact out-of-pocket figure.

✓ Instant Digital Contract Generation
Creates tamper-evident contract agreement with digital e-sign option, eliminating paper couriers and physical signature delays.

✓ Zero-Chase Real-Time Order Tracking
Customer receives SMS/WhatsApp order link with 5 live milestones (Order Placed → Feasibility → KYC → Hardware Dispatch → Line Live) and SLA alerts.

📽️ Click to Watch Demo Video (Google Drive)

📈 Measurable Business Impact & ROI

Metric	Baseline	Target	Delta
Customer Acquisition Cost	₹15,000	₹2,450	-83.7%
Lead-to-Activation Latency	17.6 Days	< 24 Hours	-95.0%
Funnel Conversion Rate	7.36%	28.0%	+3.8x
Status Inquiries per Deal	3.0 Calls	0 Calls (Self-Serve)	-100%
6-Month Post-Sale Churn	11.2%	< 4.0%	-64.3%
Activated SMBs / 5k Cohort	368 MSMEs	1,400 MSMEs	+1,032 MSMEs

💰 Annualized Value Creation for Airtel:

- CAC Savings: (₹15,000 - ₹2,450) × 1,400 activations = ₹1.75 Cr per 5,000 leads.
- Recurring ARR Boost: +1,032 incremental SMBs @ ₹2,500/mo avg = ₹3.10 Cr ARR.
- Scaled to 100,000 Annual Leads: ₹35+ Cr CAC reduction + ₹62+ Cr added ARR.

🗓️ 90-Day Execution Roadmap & Rollout

🚩 Days 1–30: Pilot in 5 Metro Circles

Deploy in Mumbai, Delhi-NCR, Bengaluru, Hyderabad, and Chennai. Train 150 Airtel Partner RMs; onboard first 1,000 SMBs; refine OCR models.

🚩 Days 31–60: Tier-2 Circle Expansion

Scale across 25 Tier-2 cities (Pune, Ahmedabad, Jaipur, Lucknow, Kochi, etc.). Integrate with Airtel BSS billing engine and CRM API.

🚩 Days 61–90: Pan-India General Rollout

Activate across all 22 Airtel telecom circles. Launch customer self-onboarding web portal alongside partner assisted channel.

📽️ REVIEWER ACTION: WATCH PRODUCT DEMONSTRATION

▶ Click here to view the official Google Drive Video Demonstration

Zero paper. Zero manual delays. Making MSMEs go digital in one sitting.